INCH POUND

MIL-C-3098/107C
5 May 1993
SUPERSEDING
MIL-C-3098/107B
8 March 1983

MILITARY SPECIFICATION SHEET

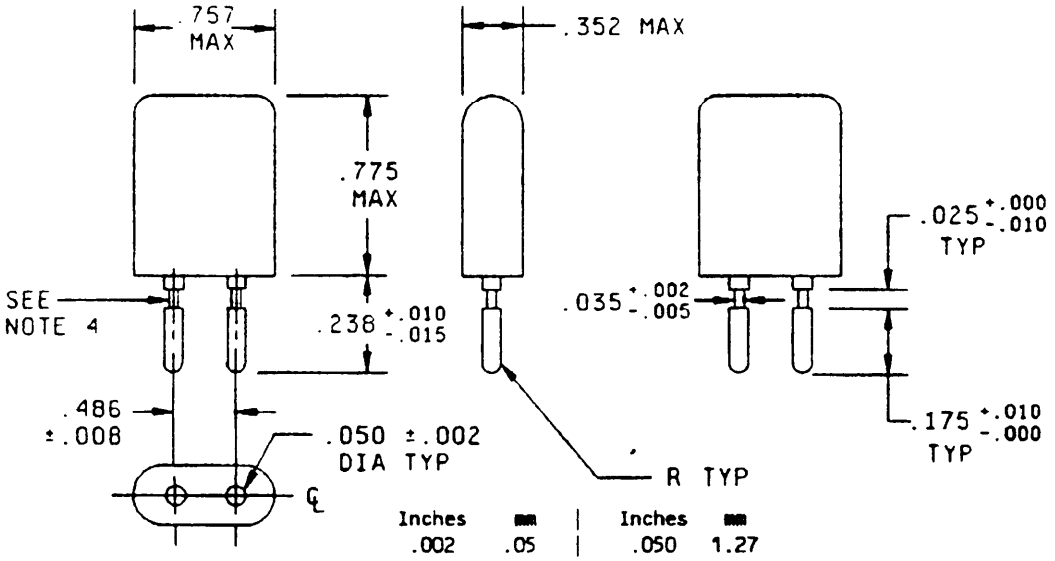
CRYSTAL UNIT, QUARTZ, CR126/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirement for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-3098.

INACTIVE FOR NEW DESIGN

Pertinent characteristics: 5 to 10 MHz; third overtone, controlled; antiresonance.




| Inches | mm | Inches | mm |
|---|---|---|---|
| .002 | .05 | .050 | 1.27 |
| .005 | .13 | .175 | 4.45 |
| .008 | .20 | .238 | 6.05 |
| .010 | .25 | .352 | 8.94 |
| .015 | .38 | .486 | 12.34 |
| .025 | .64 | .757 | 19.23 |
| .035 | .89 | .775 | 19.69 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-C-3098.
4. The pin undercut may be omitted.

FIGURE 1. Crystal unit - CR126/U.


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REQUIREMENTS:

Holder: HC-27/U (MIL-H-10056/14).

Dimensions, marking, and configuration: See figure 1.

Frequency range: 5 to 10 MHz, inclusive.

Frequency tolerance: 14 ohms maximum.

Equivalent resistance: -3.5 ppm +1.5 ppm.

Antiresonance, load capacitance: 30 ±0.5 pF.

Mode of oscillation: Third overtone.

Operable temperature range: -55°C to +90°C, inclusive.

Operating temperature range: +81°C to +89°C.

Crystal unit quality (Q): $10^6$ minimum. 1/

Rated drive level: 2 ±1 mW.

Capacitance shunt: 7 pF, maximum.

Shock (specified pulse):

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Vibration: Method 204 of MIL-STD-202, test condition A.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

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1/ The crystal unit quality shall be not less than the value specified, as determined during measurement, using the following formula:

$$Q = \frac{1}{4\pi(\Delta f)(C_T)(R)}$$

Where: $\Delta f$ = difference, in Hz, between the series and antiresonant frequency at $C_L$ (load capacitance) of 30 pF.
$C_T$ = Capacitance, $C_0$ (shunt capacitance of the crystal unit) plus $C_L$ (load capacitance) of 30 pF.
$R$ = Motional-arm resistance of the crystal unit.



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QUALITY ASSURANCE PROVISIONS:

Qualification inspection: Not applicable for this specification.

Quality conformance inspection: Group A tests of MIL-C-3098 shall be applicable.

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI, SM
Navy - SH
Air Force - 99
DLA - ES

User activities:
Navy - AS, CG, MC, OS
Air Force - 17, 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5955-0678-04)



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